



第十四章

智能图像彩色化处理



01 智能图像彩色化技术的发展

02 基于交互的图像上色

03 基于多模态的自动图像上色



01

智能图像彩色化技术
的发展

智能图像彩色化技术的发展

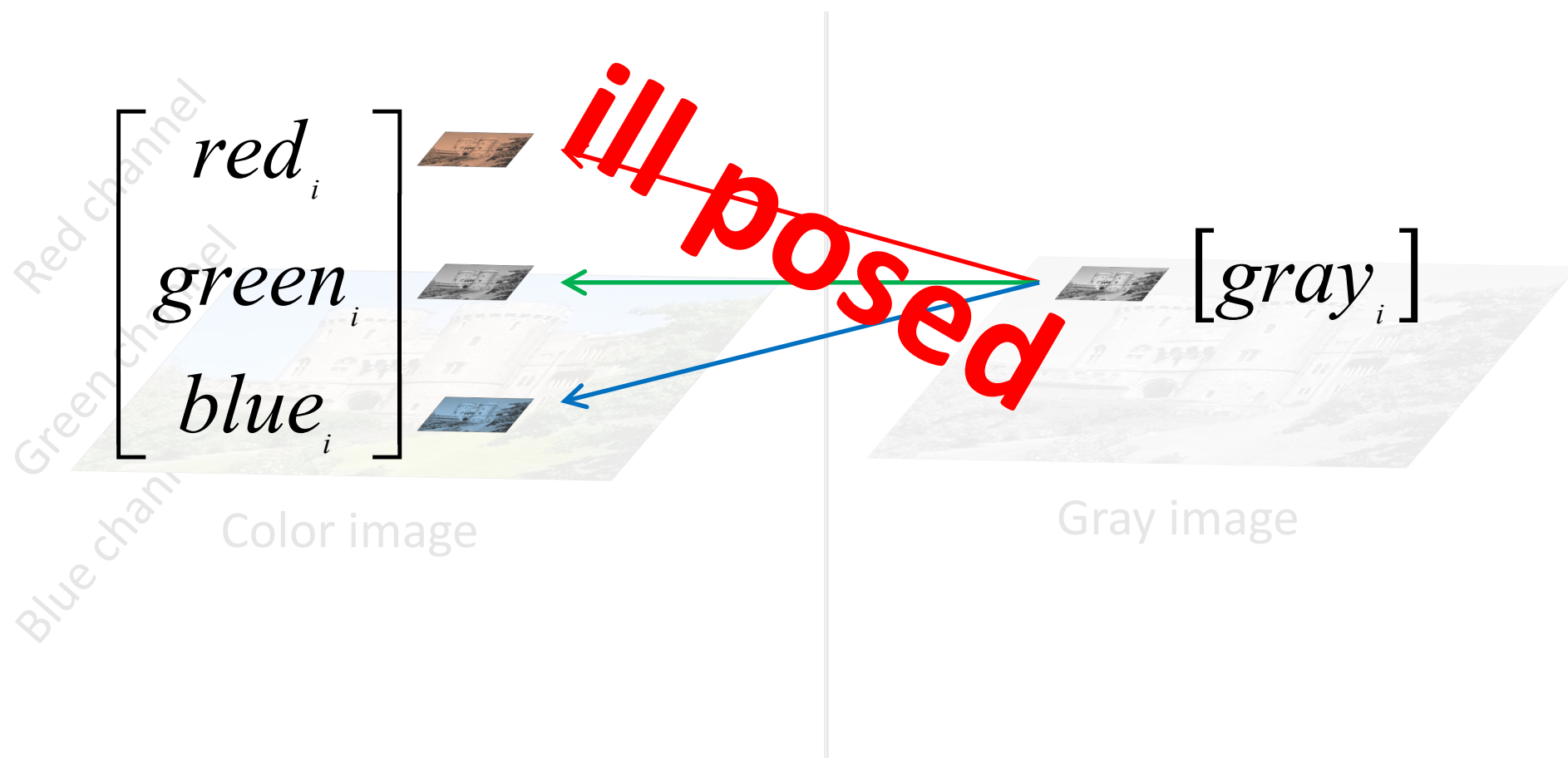
图像彩色化(image colorization):定义

The aim of colorization is to convert a **grayscale image** to **color image**.



智能图像彩色化技术的发展

图像彩色化(image colorization): 挑战



图像彩色化(image colorization)分类

◆ 用户交互图像上色

- 基于颜色提示的上色
- 基于彩色参考图的上色
- 基于语言/文字的上色

◆ 基于多模态的自动图像上色

智能图像彩色化技术的发展

图像彩色化(image colorization): 应用

- ✓ 旧照片或旧电影彩色化处理
- ✓ 夜视仪图像实时彩色化
- ✓ 漫画上色



智能图像彩色化技术的发展

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- ✓ 旧照片或旧电影彩色化处理
- ✓ 夜视仪图像实时彩色化
- ✓ 漫画上色



SPI X27全彩夜视仪

智能图像彩色化技术的发展

图像彩色化(image colorization): 应用



机场X光成像

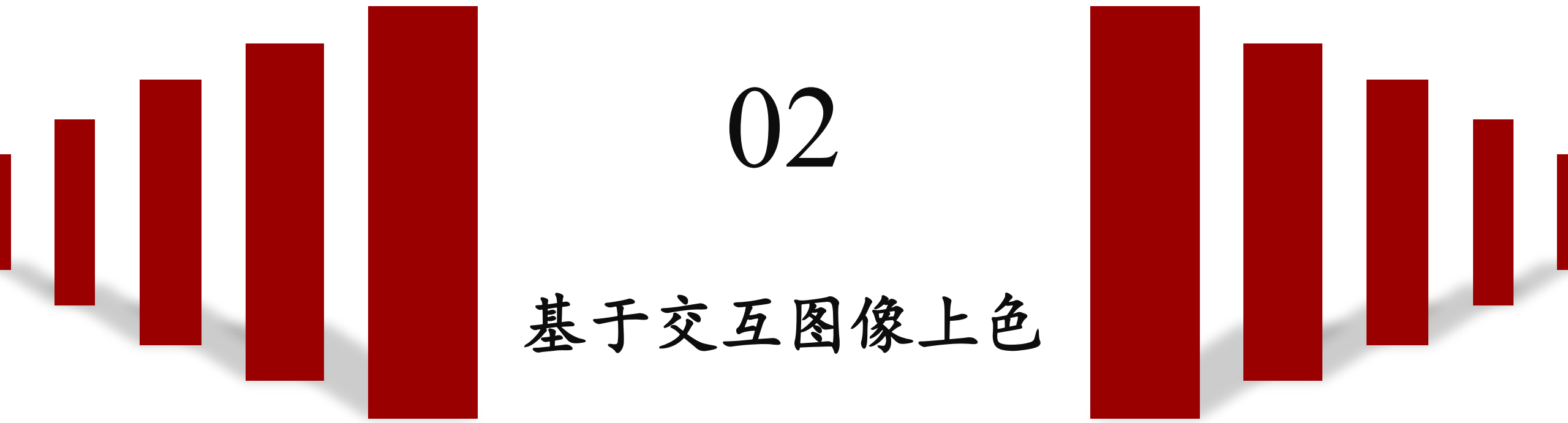
纸张、衣服和大多数爆炸物在扫描时呈现橙色，铝则显示为绿色，而铜则以蓝色呈现。若材料无法穿透，则显示为黑色。

双能X射线成像技术：通过交替使用两种不同的X射线波长来扫描物体，并综合两幅图像的信息来分析物体中的物质成分。



CT：突出显示类似痛风的结构和松质骨

痛风，一种关节周围晶体沉积物形成的疾病，其特性使得痛风晶体在双能CT扫描中呈现绿色，可直观地反映痛风治疗的效果。此外，紫色在扫描中代表松质骨。



02

基于交互图像上色

◆用户交互图像上色

- 基于颜色提示的上色
- 基于彩色参考图的上色
- 基于语言/文字的上色

黑白草稿图上色的挑战

1. 由于素描的主题和绘画风格不受限制，因此很难自动理解草图。

2. 素描通常以简化抽象的形式表达，或者很少有平滑的明暗处理来指导着色。



colorization



基于交互图像上色

基于颜色提示的上色



只需要输入黑白图像、和图像上对应部位的颜色点 或 线条 或 涂鸦，即可得到彩色化后的图像。

Online Demo: https://petalica-paint.pixiv.dev/index_zh.html

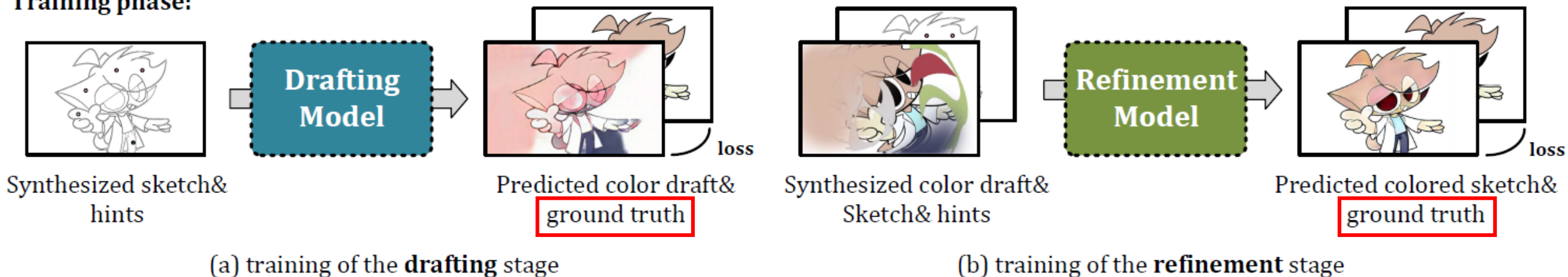
基于颜色提示的上色



Fig. 1. Our framework colorizes a sketch with **two stages**. The first stage renders the sketch into a rough color draft image. To refine the color draft, we introduce a second stage to identify mistakes and refine them, in order to obtain the final result. © AU (DeviantArt artist).

模型训练流程

Training phase:



(a) 绘图阶段，通过训练来确定颜色的**构成**和**区域**，并预测出颜色的草图。

(b) 细化阶段：识别不正确的颜色区域，并使用一组附加的**用户提示**对其进行细化。

模型测试流程



Fig. 3. **Overview of our framework**, including data preparation, the training and testing pipeline. © Senor-G (mouse), style2paints (girl).

◆ 用户交互图像上色

◆ 基于颜色提示的上色

➤ 基于彩色参考图的上色

◆ 基于颜色色块的上色

◆ 基于语言/文字的上色

基于交互图像上色

基于彩色参考图的上色



Input image

+



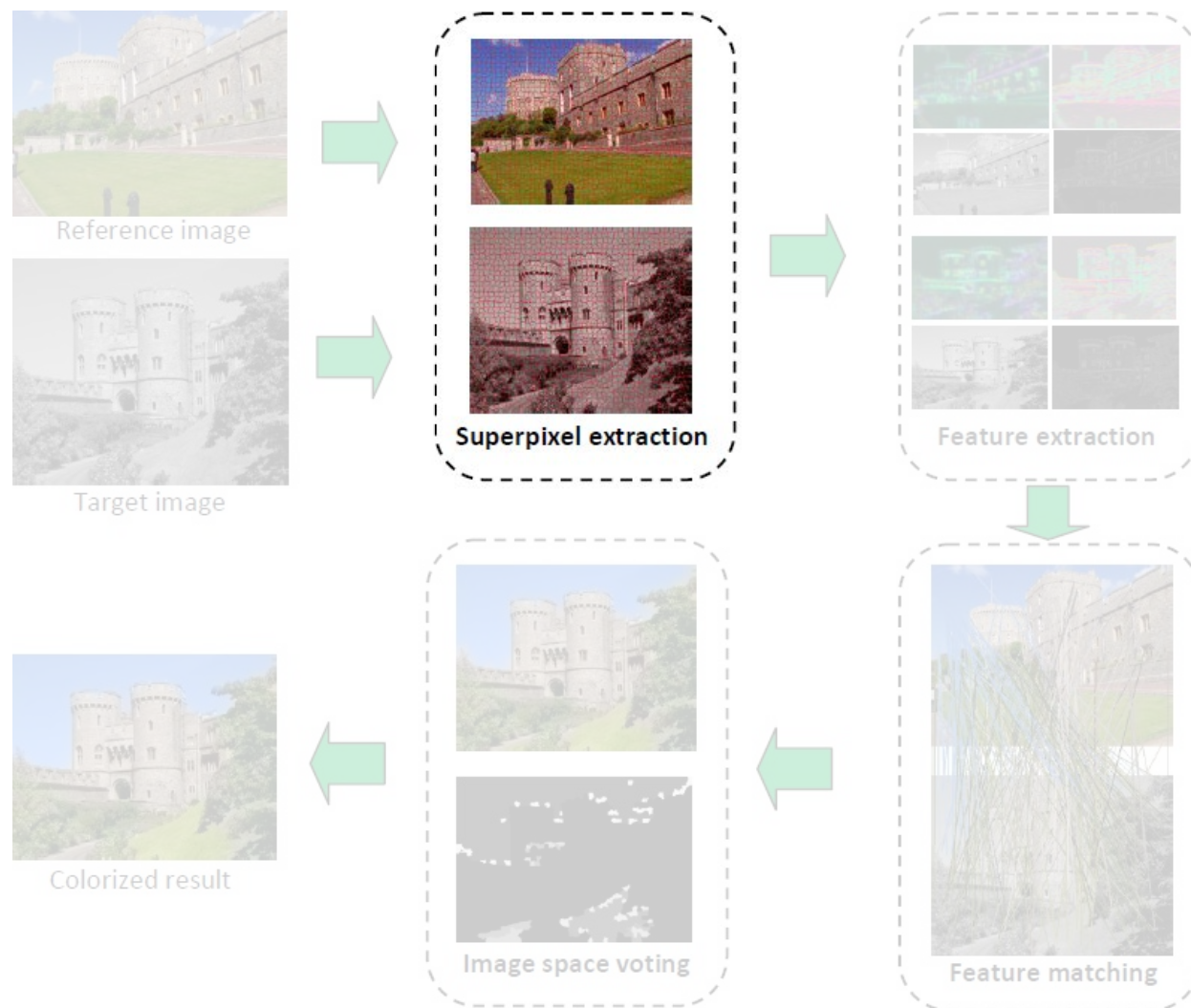
Example image

=

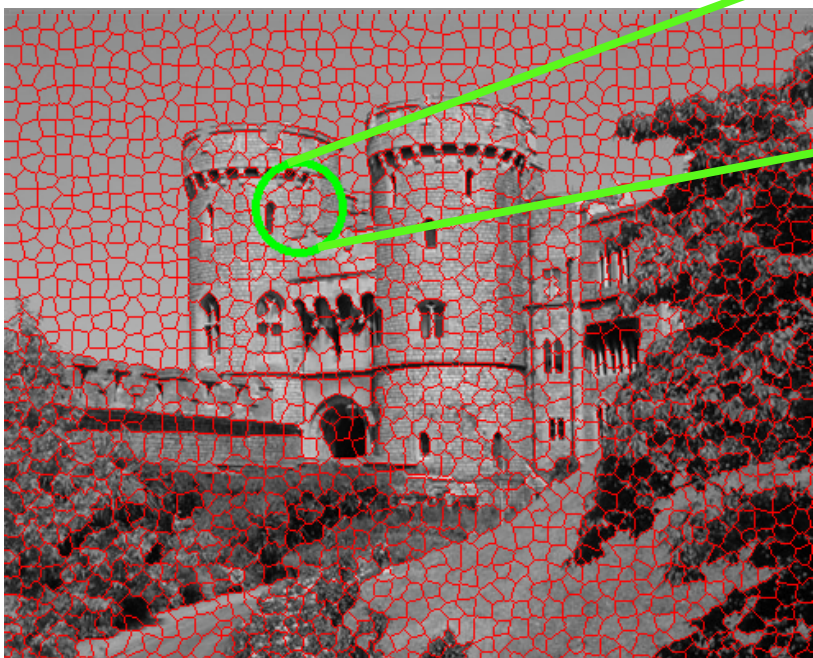


Colorized Result

基于彩色参考图的上色: System overview

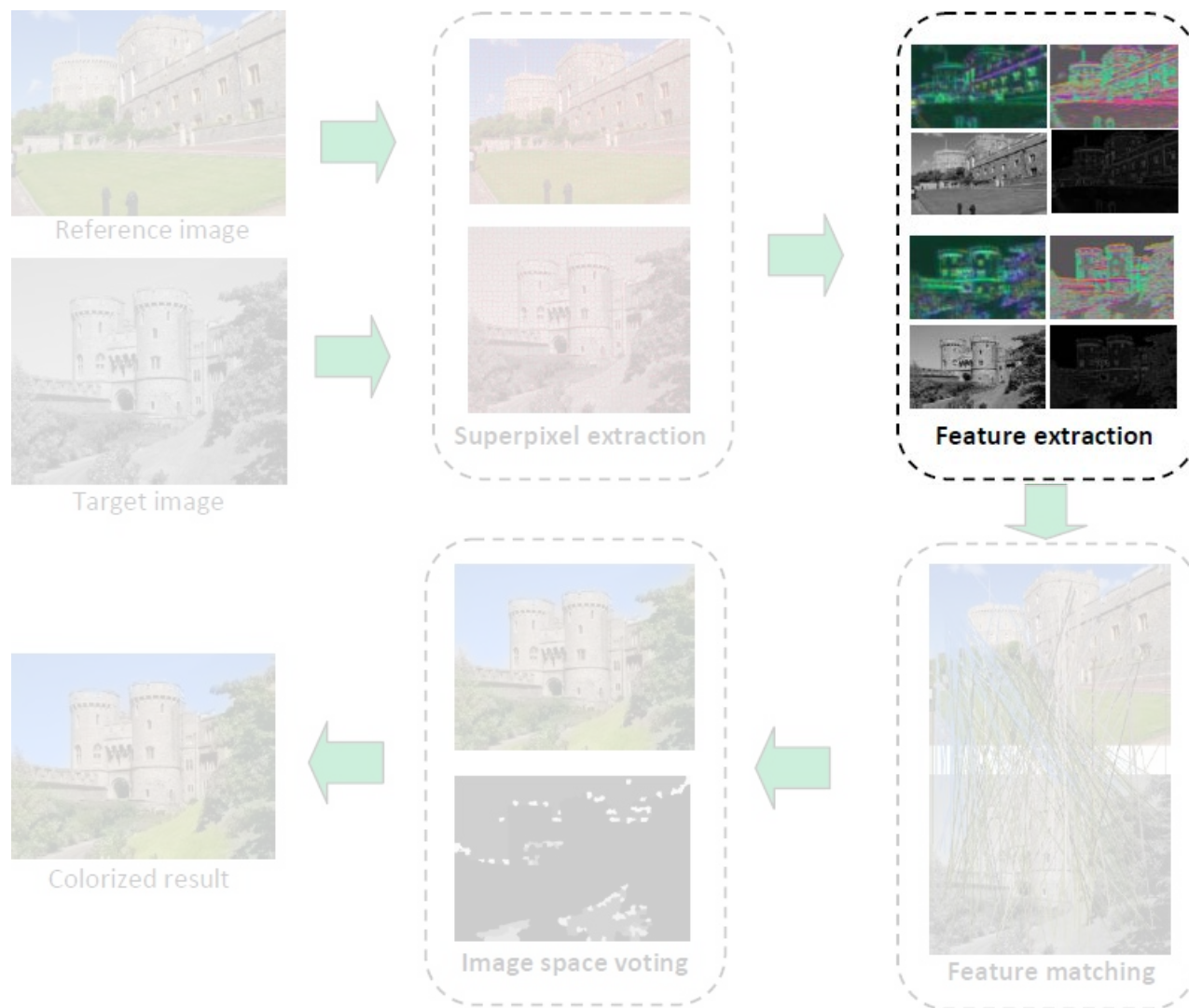


基于彩色参考图的上色: Superpixel Extraction

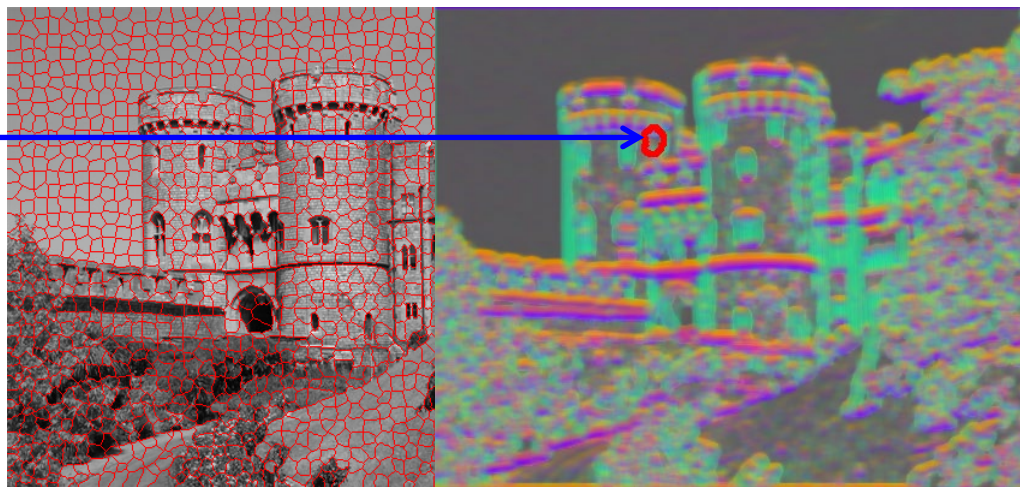


- Reduces complexity by grouping pixels.
- Uniform size and coverage.
- Compactness.
- No overlap.

基于彩色参考图的上色：System overview



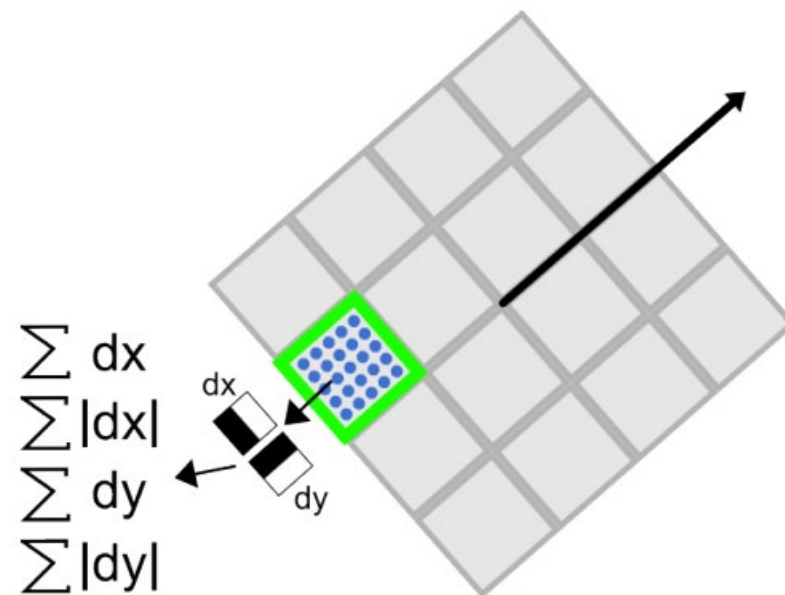
基于彩色参考图的上色: Feature Extraction



SURF Features

$$D_s = \{d_1, d_2, d_3, \dots, d_{128}\}$$

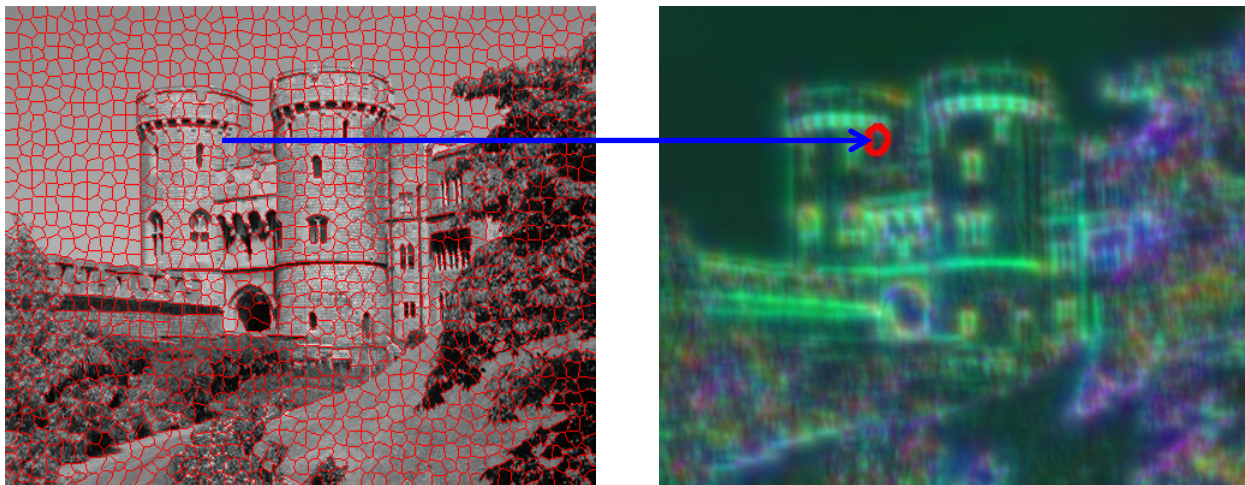
Image Features : **64D SURF Features**



64D SURF Features

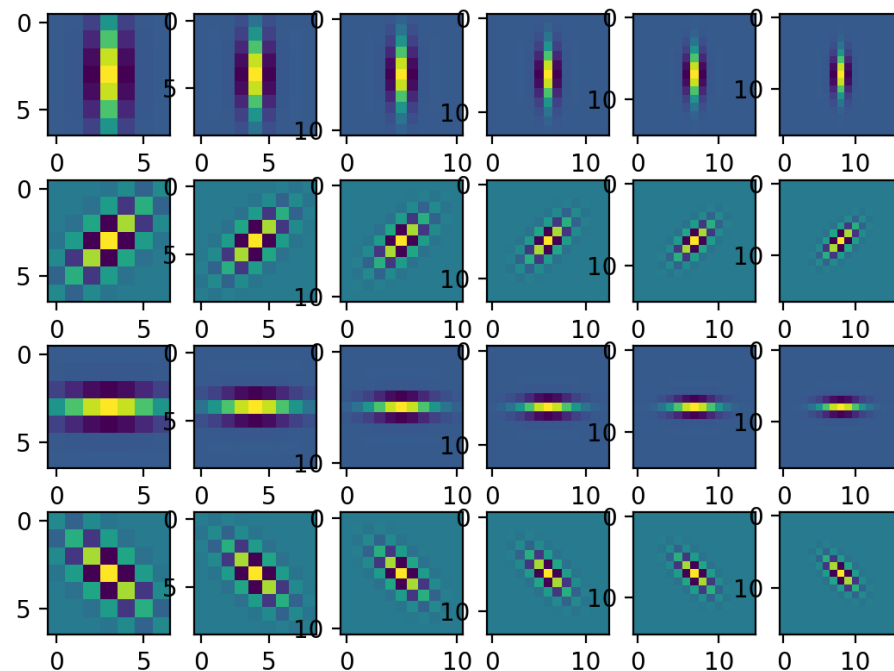
4×4 (小区域- 5×5 个像素) $\times$ 4 (haar小波特征)

基于彩色参考图的上色: Feature Extraction



Gabor Features

Image Features: **40D Gabor Features**
(5 scales $\times$ 8 orientations)



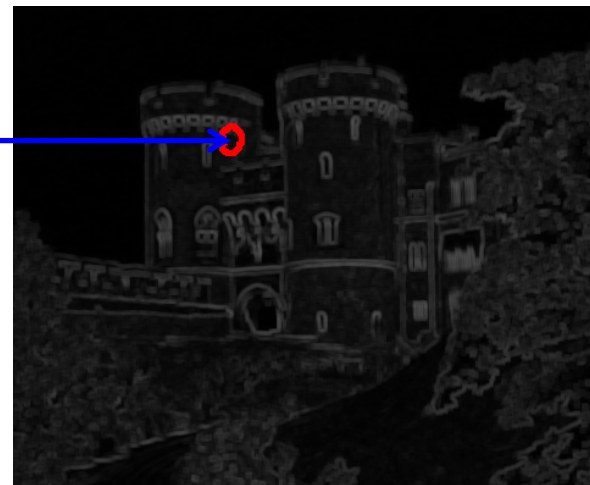
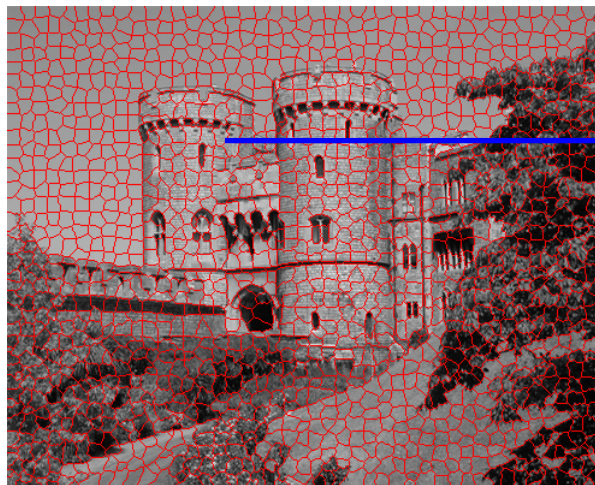
6 scales $\times$ 4 orientations
6 scales: 7、9、11、13、15、17

Gabor feature : 一般是通过对图像与gabor filter做卷积而得到;

Gabor filter: 定义为高斯函数与正弦函数的乘积。

Gabor是一个用于边缘提取的线性滤波器, 其频率和方向表达与人类视觉系统类似, 能够提供良好的方向选择和尺度选择特性, 而且对于光照变化不敏感, 因此十分适合纹理分析

基于彩色参考图的上色: Feature Extraction



Standard Deviation

Image Features : **2D Std. Dev. Features**

基于彩色参考图的上色: Feature Extraction



Standard Deviation

$$D_{sd} = \{d_1, d_2\}$$

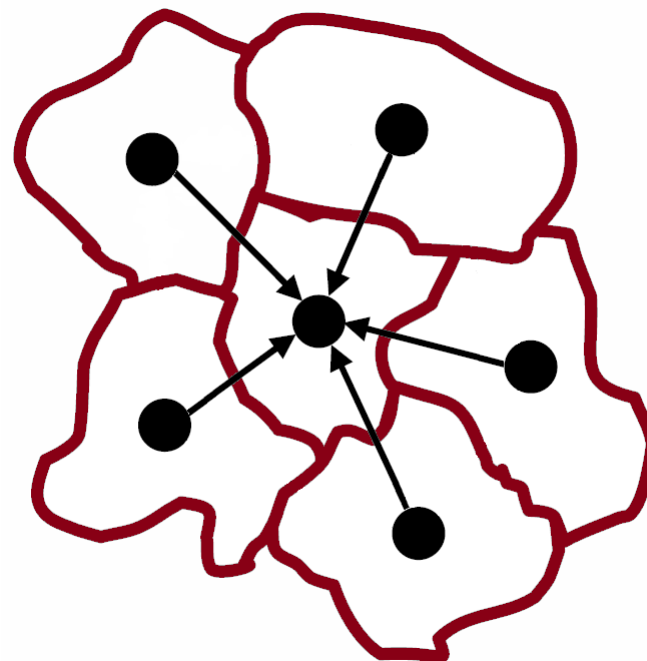
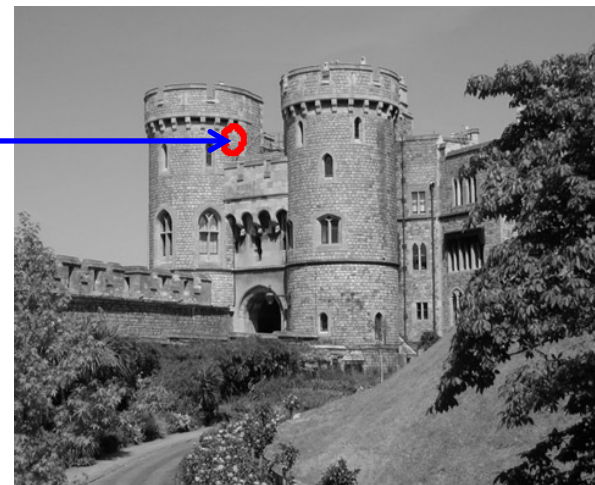


Image Features : **2D Std. Dev. Features**

基于彩色参考图的上色: Feature Extraction



Intensity

Image Features : **2D Intensity features**(Similar to Standard Deviation)

基于彩色参考图的上色: Color Quantization

- **CIELab** color space is used to discretize the color values.

Supapixel	Color Value (C)
	(a_1, b_1)
	(a_2, b_2)
	(a_3, b_3)
• • •	• • •
	(a_{n-1}, b_{n-1})
	(a_n, b_n)

Lab模式，三个通道：

- 第一个通道是明度，即“L”。
- a通道：颜色是从红色到深绿；
- b通道：则是从蓝色到黄色。

基于彩色参考图的上色: Color Quantization

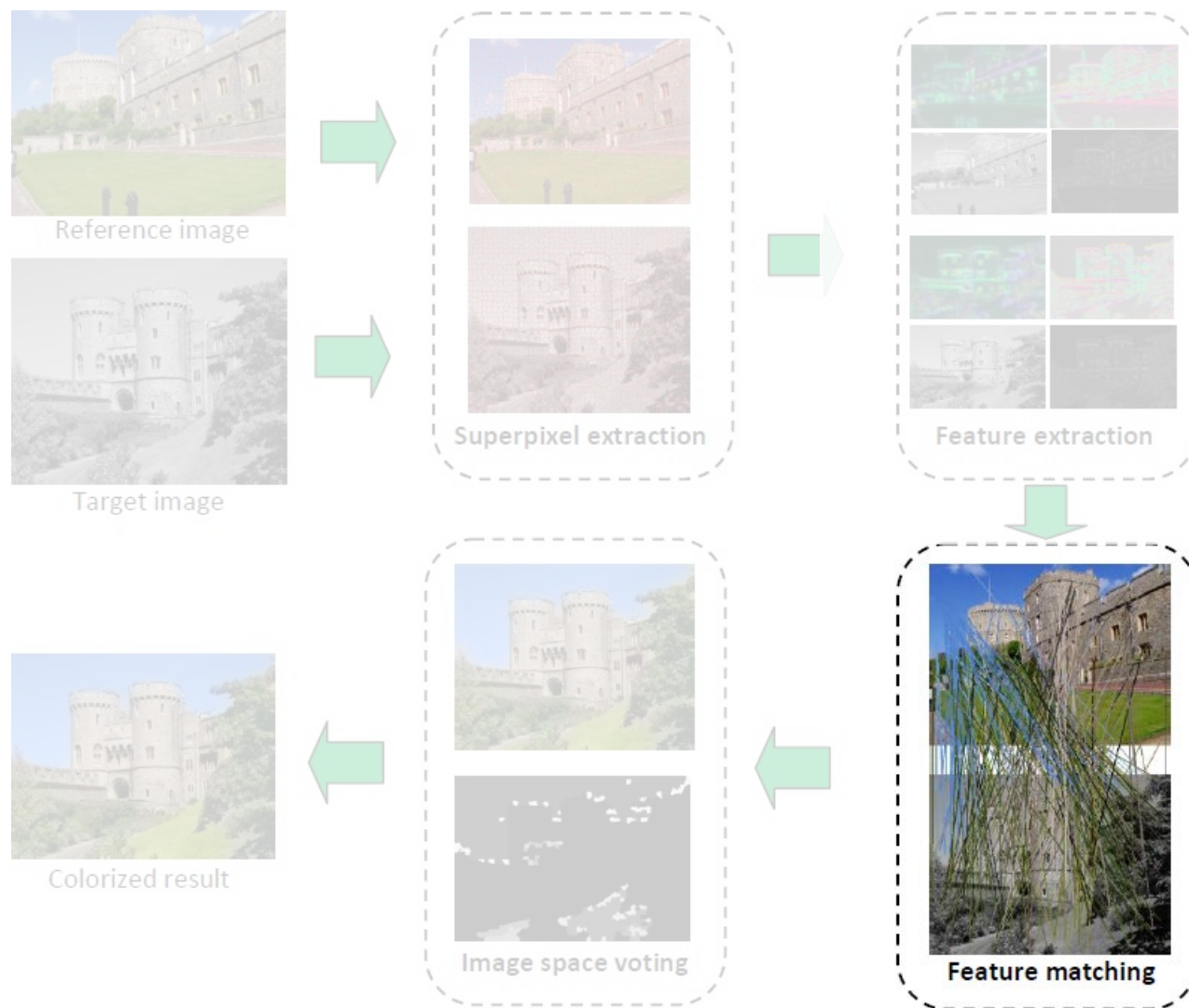
- ***k*-means** clustering has been used to group the chromatic color values.

Superspixel	Color Label (C)
	1
	
	
• • •	• • •
	k
	

基于彩色参考图的上色: Color Quantization

Supapixel	Features	Color Label (C)
	$f_{surf}, f_{gabor}, f_{sd}, f_{int}$	1
	$f_{surf}, f_{gabor}, f_{sd}, f_{int}$	
	$f_{surf}, f_{gabor}, f_{sd}, f_{int}$	
• • •	• • •	• • •
	$f_{surf}, f_{gabor}, f_{sd}, f_{int}$	k
	$f_{surf}, f_{gabor}, f_{sd}, f_{int}$	

基于彩色参考图的上色：System overview



基于彩色参考图的上色: **Feattrue matching**

- **Cascade feature** matching scheme.
- Search space are **pruned** at each step of the cascade.
- Discriminative Gabor and SURF features are used at the initial cascade steps.

$$f_{surf} \longrightarrow f_{gabor} \longrightarrow f_{sd} \longrightarrow f_{intensity}$$

基于彩色参考图的上色: Feature matching

- The matching cost is computed as:

$$F(r_b, t_i) = w_1 M_1(r_b, t_i) + w_2 M_2(r_b, t_i) + w_3 M_3(r_b, t_i) + w_4 M_4(r_b, t_i)$$

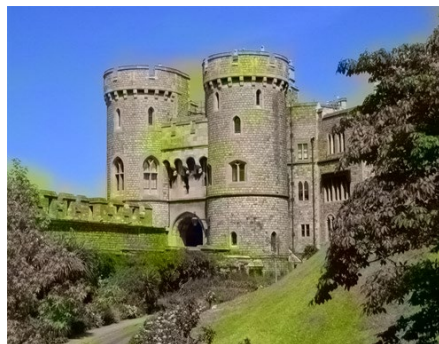
- M_1 , M_2 , M_3 and M_4 are the Euclidean distance between the **Gabor**, **SURF**, **intensity** and **standard deviations** features, and
- w_1 , w_2 , w_3 and w_4 are their accompanying weights fixed as 0.2, 0.5, 0.2 and 0.1, respectively.

基于交互图像上色

基于彩色参考图的上色:
Featruue matching



Gabor



SURF



Intensity



Std. Dev.



Gabor



Gabor+SURF

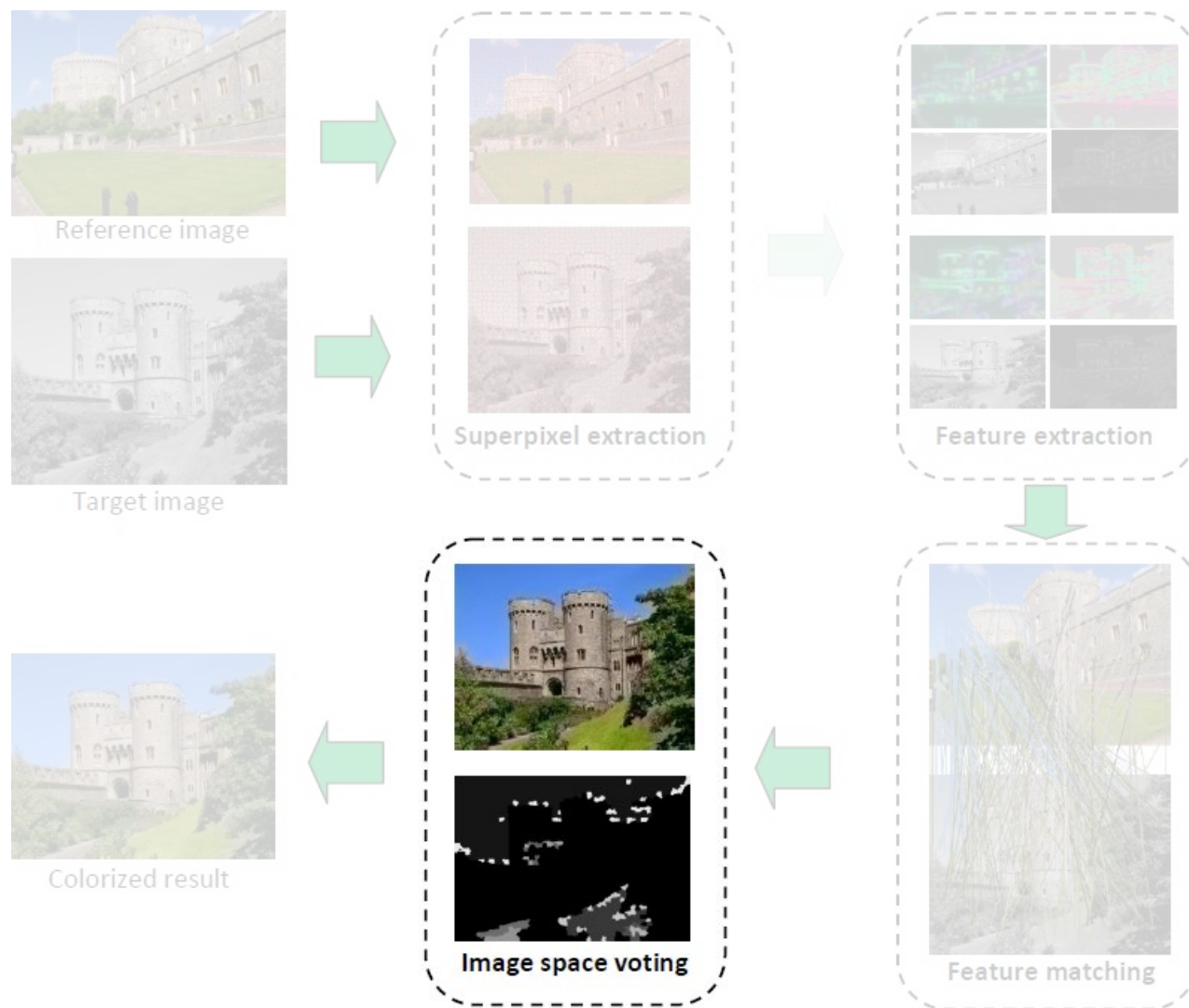


Gabor+SURF+
Intensity

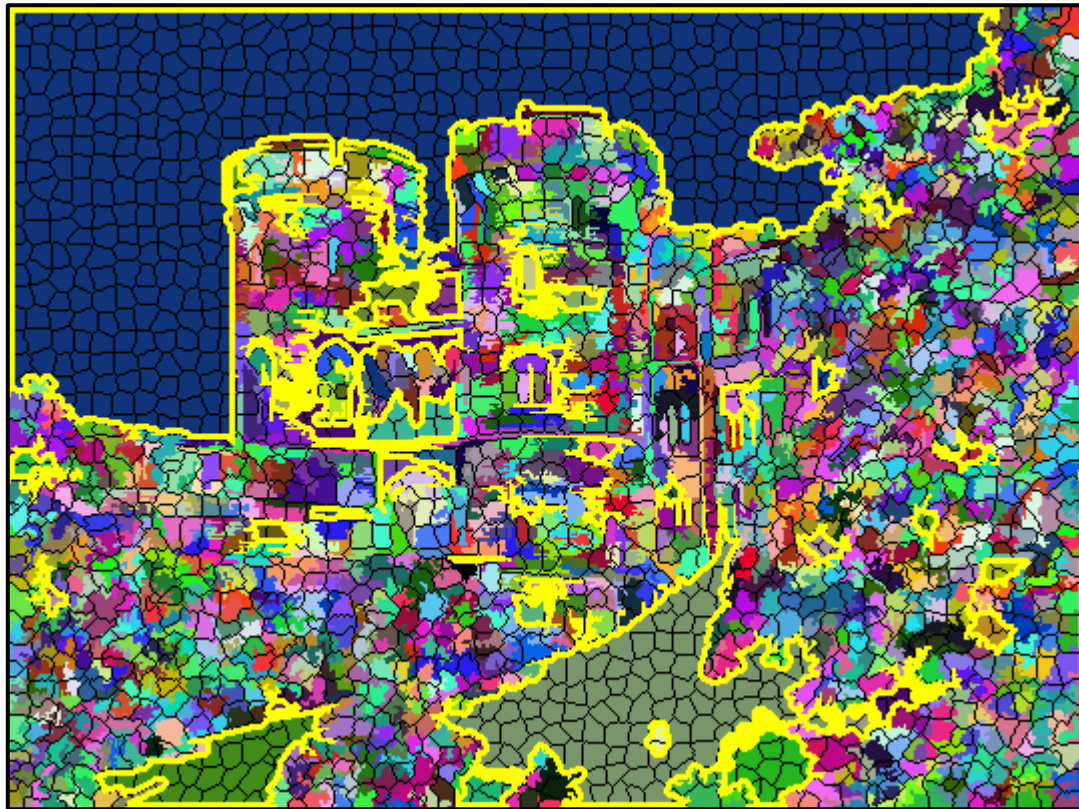


Gabor+SURF+Intensity
Std. Dev.

基于彩色参考图的上色：System overview



基于彩色参考图的上色: Color Scribble Refinement



为细化颜色涂鸦,

- 1) 首先, 执行图像分割, 得到分割图斑, 即**超像素**
- 2) 对**超像素分组**, 并执行基于颜色标签的投票。
 - ① 投票, 只使用在显示的图像中至少有3个超像素的片段。
 - ② 将具有最大投票数的颜色标签指定给该图像区域中的所有超像素。

基于彩色参考图的上色: Color Scribble Refinement

- Color Scribbles are propagated to neighboring pixels with similar luminance values through optimization.



基于彩色参考图的上色: Results



Input Gray Image



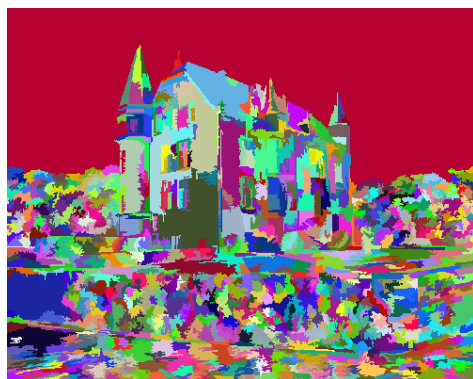
Reference Image



Colorization
Result

基于彩色参考图的上色: 挑战

- Superpixel accuracy at object boundaries.
- Availability of suitable color exemplars.
- Small image segments in dense textured image regions.



◆ 用户交互图像上色

- 基于颜色提示的上色
- 基于彩色参考图的上色
- 基于语言/文字的上色

基于交互图像上色

基于语言/文字的上色

给定一个草图，计算机接受和分析用户输入语言指令并自动地将草图变成彩色卡通图。

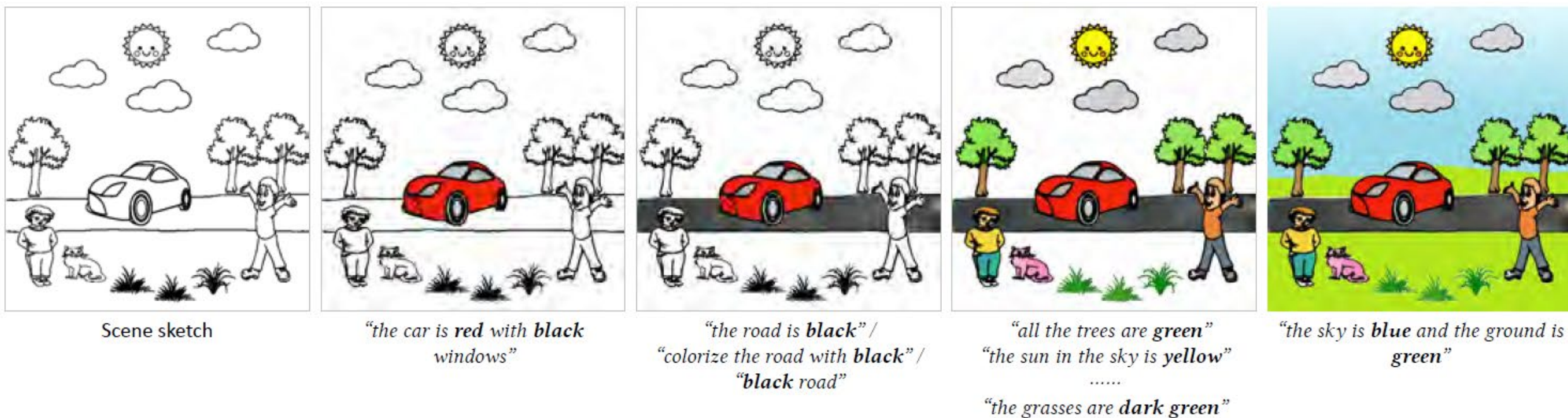


Fig. 1. Given a scene sketch, our system automatically produces a colored cartoon image by progressively coloring foreground object instances and the background following user-specified language-based instructions.

基于交互图像上色

基于语言/文字的上色：挑战

涉及到两种非常复杂的不同模态输入数据（自然语言和草图）的理解。



*the bus has orange body and
blue windows*

(a)



*the two trees on the left of the
house are light green*

(b)



all the clouds are dark gray

(c)

基于语言/文字的上色：解决方法

将问题解耦为分割、涂色**两个子问题**：

1 分割：根据用户的文字输入自动分割出感兴趣的物体；

2 涂色：基于生成对抗网络(GAN)以及图-文特征融合算法的模型根据用户指令对分割出的物体进行上色。

基于语言/文字的上色：**缺点**

- 自然语言比较难区分相近的颜色
- 对用户指令理解泛化能力不足



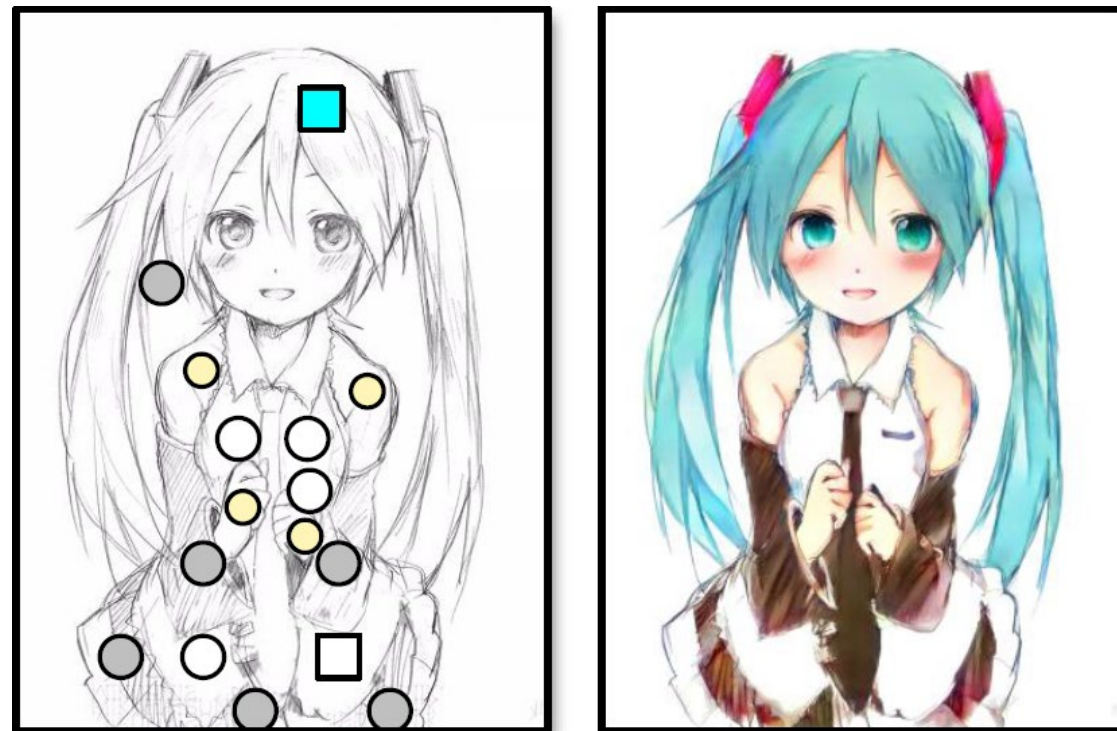
03

多模态的自动图像 上色

问题/挑战:

- The scribble-based methods typically require **substantial efforts** from the user to **provide considerable scribbles** on the target grayscale images.

It is **time-assuming** to colorize a grayscale image with finescale structures, especially for a rookie user.

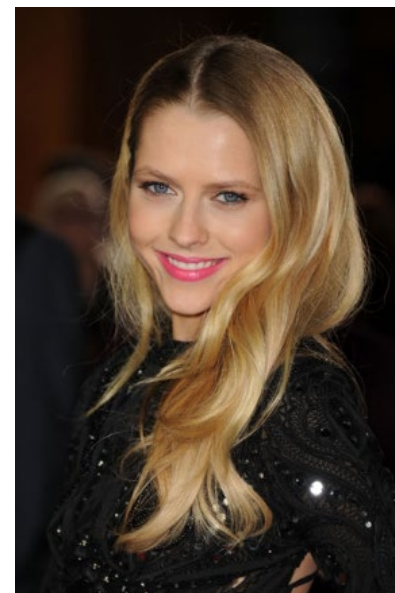


多模态的自动图像上色

- To reduce the burden on user, **example-based method** typically transfers the color information from a **similar reference** image to the target grayscale image.



Input Gray Image



Reference Image



Colorization
Result



Hard to finding a suitable reference image!

多模态的自动图像上色

- 用户提示的图像上色: 用户告诉计算机, 图片中某个对象应该是什么颜色。
- 全自动图像上色: 把一个巨大的彩色图像数据集喂给模型, 让模型学习图像中每个对象的颜色。

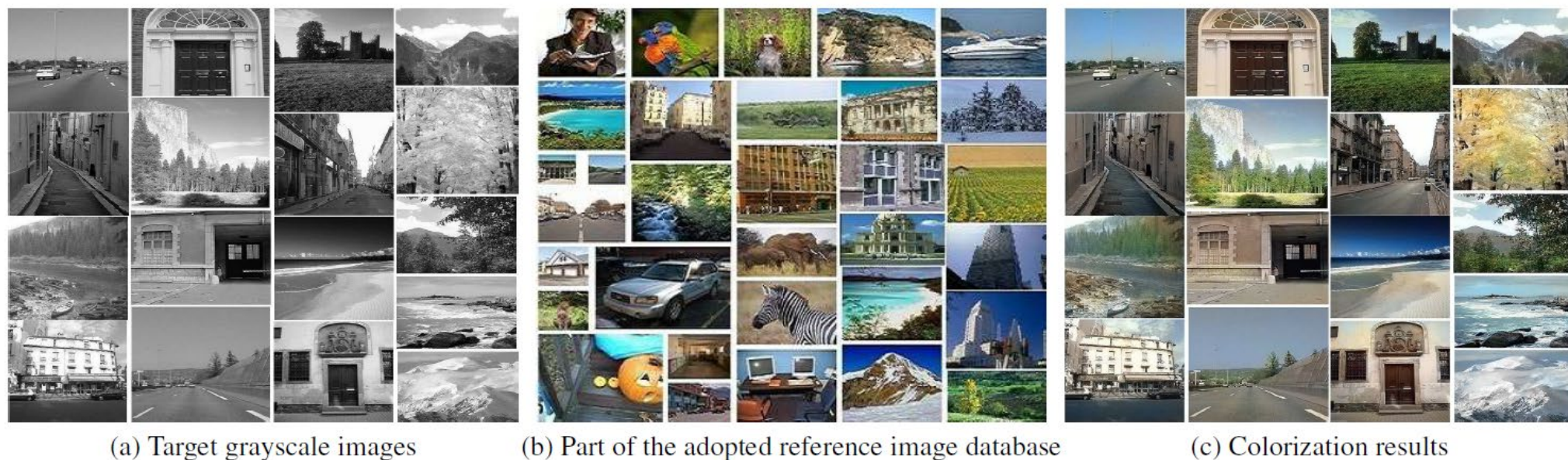


Figure 1. The colorization results of our full-automatic method. Our system utilizes a large database of colorful reference images as shown in (b). After the training of neural networks, the learned model is used directly to colorize the target gray scale images in (a). The colorization results are presented in (c).

多模态的自动图像上色

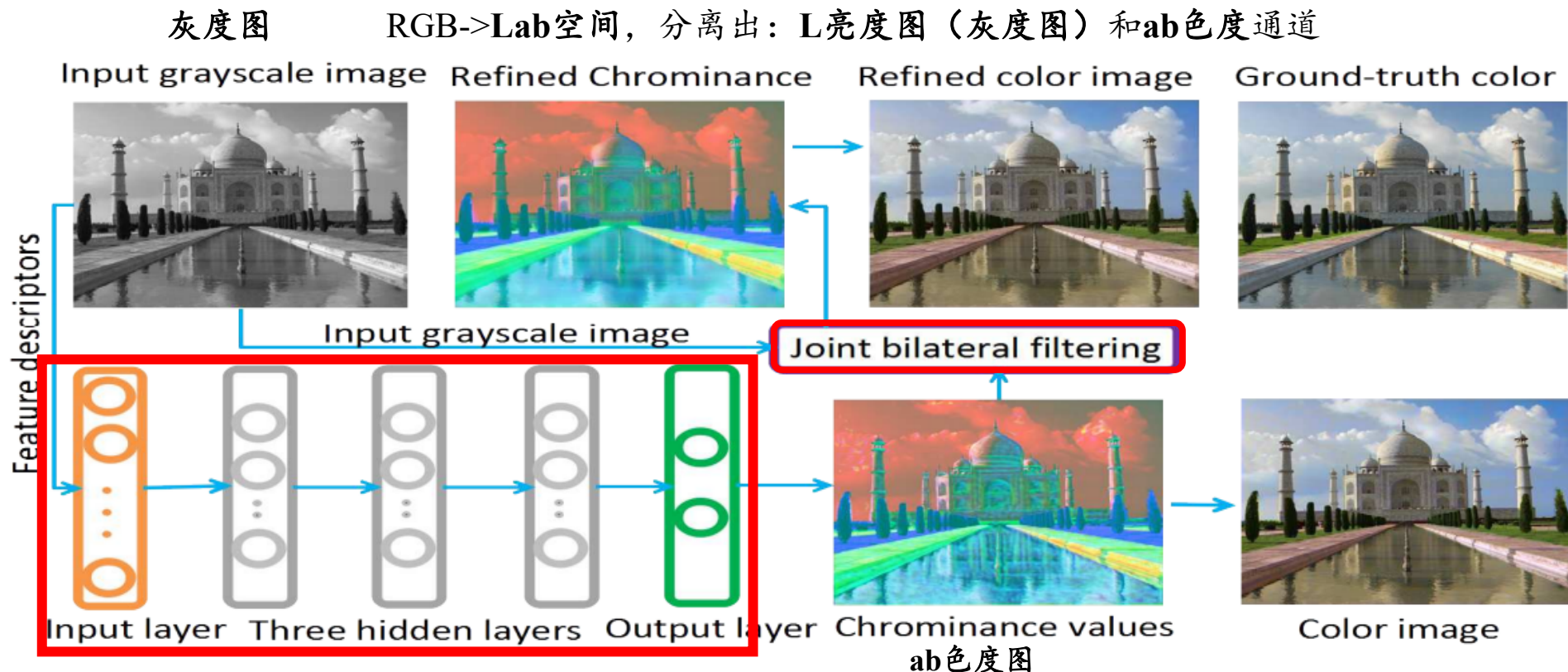


Figure 2. Overview of the proposed colorization method and the architecture of the adopted deep neural network. The feature descriptors will be extracted at each pixel and serve as the input of the neural network. Each connection between pairs of neurons is associated with a weight to be learned from a large reference image database. The output is the chrominance of the corresponding pixel which can be directly combined with the luminance (grayscale pixel value) to obtain the corresponding color value. The chrominance computed from the trained model is likely to be a bit noisy around low-texture regions. The noise can be significantly reduced with a joint bilateral filter (with the input grayscale image as the guidance).

如何实现自动图像上色？

步骤一：使用较大的数据集训练一个神经网络

Algorithm 1 Image Colorization – Training Step

Input: Pairs of reference images: $\Lambda = \{\vec{G}, \vec{C}\}$.

Output: A trained neural network.

1. Compute feature descriptors $\vec{x}$ at sampled pixels in $\vec{G}$ ——灰度图像
and the corresponding chrominance values $\vec{y}$ in $\vec{C}$; ——彩色图像
 2. Construct a deep neural network;
 3. Train the deep neural network using the training set Ψ
 $= \{\vec{x}, \vec{y}\}$. {特征描述符、色度值}
-

如何实现自动图像上色？

步骤二：把黑白图像输入到训练好的神经网络-预测+优化

Algorithm 2 Image Colorization – Testing Step

Input: A target grayscale image I and the trained neural network.

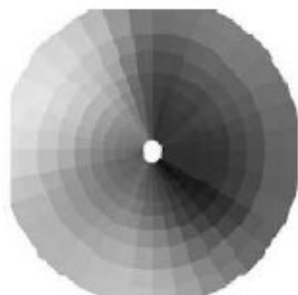
Output: A corresponding color image: $\hat{I}$.

1. Extract a feature descriptor at each pixel location in I ;
 2. Send feature descriptors extracted from I to the trained neural network to obtain the corresponding chrominance values;
 3. Refine the chrominance values to remove potential artifacts;
 4. Combine the refined chrominance values and I to obtain the color image $\hat{I}$.
-

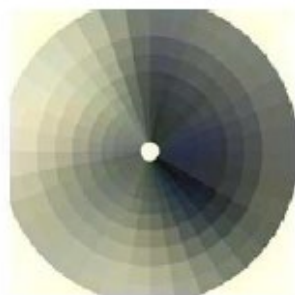
多模态的自动图像上色

自动图像上色方法的缺点

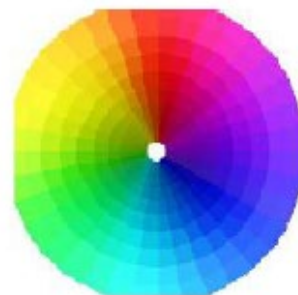
- 只能预测在训练集中出现过的对象
- 对一些合成图像预测效果不好



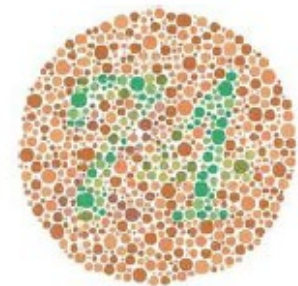
(a) Input



(b) Our method



(c) Groundtruth



多模态的自动图像上色

Automatic controllable colorization via imagination



意义和挑战:

- 大量黑白照片和历史影像资料需要被赋予色彩以更好地呈现其历史和文化价值。
- 然而，手动为这些图像上色是一项耗时且费力的工作。
- 现有自动上色存在色彩多样性和用户交互性方面仍存在不足。

解决方案:

- 提出了一种基于想象模块的自动可控上色框架。通过模拟人类专家的想象过程，该框架能够生成多种色彩丰富且结构一致的参考图像。
- 此外，该框架还支持用户交互式调整，使得上色结果更加符合用户的个性化需求。

多模态的自动图像上色

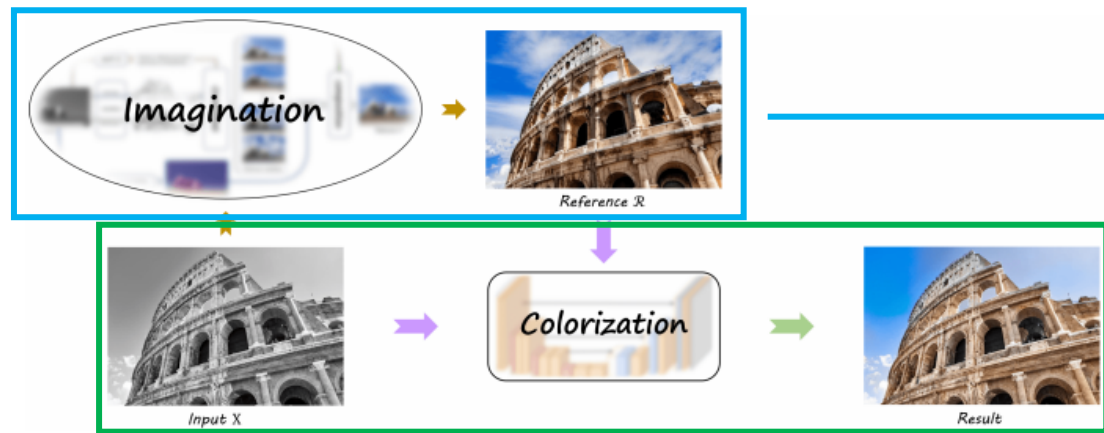


Figure 2. **Framework Overview.** Given a black-and-white input, our framework first synthesizes a semantically similar, spatially aligned, and instance-aware reference by emulating the imagination process of human experts. Then the colorization module colorizes the black-and-white image with the guidance of reference. On the right, we present an additional example.

框架包括3大模块：

- ① **想象模块(Imagination)**：根据输入的灰度图像 X ，利用预训练的可控制的图像生成模型(ControlNet)，合成 N 幅语义相似且结构一致的参考图像。这些图像作为上色的参考，模拟了人类专家的想象过程。
- ② **参考细化模块(Reference Refinement)**：通过语义分割和特征匹配等方法，从多个参考图像中合成最优的参考图像 R 。该模块能够提升上色结果的色彩丰富性和一致性。
- ③ **上色模块(Colorization)**：基于示例上色的方法，利用精炼后的参考图像 R 对黑白输入图像进行上色。通过粗到细的提示颜色生成和优化过程，实现更精准的上色效果。



Figure 5. **Diverse Colorization.** We can synthesize diverse colorful references from Imagination Module 3.3, yielding diverse colorization results.

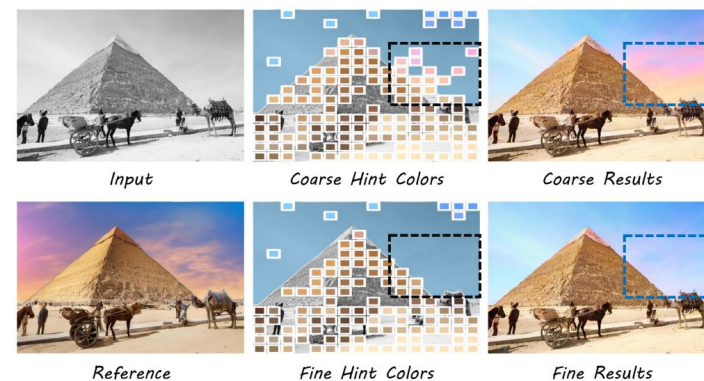
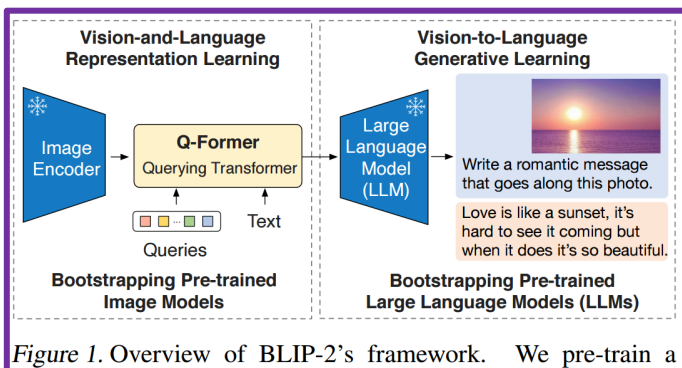


Figure 7. **Coarse-to-fine hint colors optimization** can mitigate the color ambiguity within semantic instances by identifying the most representative hint colors, as highlighted by the boxed area in the figure.

多模态的自动图像上色



Semantic-SAM

Figure 3. **Imagination Module and Reference Refinement Module.** In **Imagination Module 3.3**, given a grayscale input, we generate N reference candidates C , N_1 of which are conditioned on the canny edge of X , and N_2 of which are conditioned on the HED boundary of X , $N = N_1 + N_2$. In **Reference Refinement Module 3.4**, we first extract the segmentation S of X . For each segment S^j , we look for the best reference segment $R \odot S^j$ for the optimal reference R by selecting the nearest neighbor for $X \odot S^j$ among all reference candidates $C_i \odot S^j$ in terms of the difference in the robust and universe DINOv2 [53] feature space.

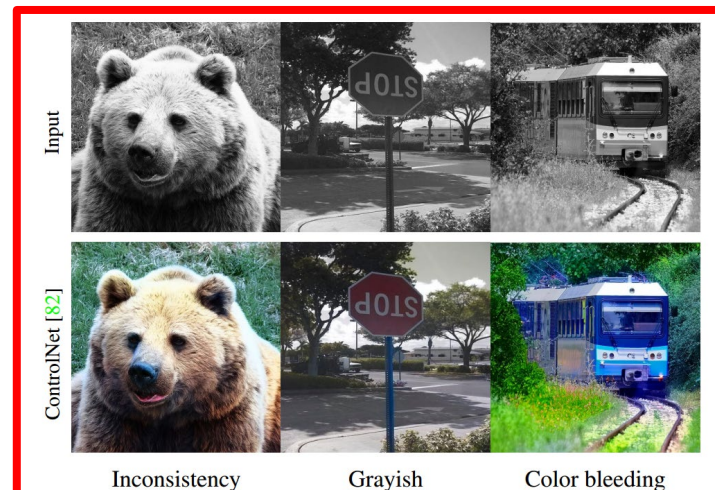


Figure 4. **ControlNet Issues.** Directly applying ControlNet[82] to colorization might cause issues like inconsistency, grayish, and

ControlNet通过在Stable Diffusion模型中添加辅助模块，从而引入“额外条件”来控制AI绘画的生成过程。

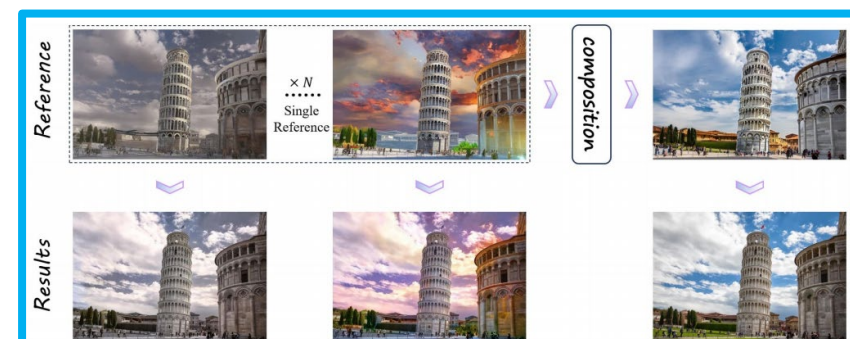


Figure 11. **Ablation Study: Reference Refinement Module 3.4.** Without our composition strategy, the diversity and completeness of the color space are compromised. Consequently, the reference exhibits inconsistency, visual artifacts, and color bleeding, leading to unsatisfactory colorization results.

多模态的自动图像上色

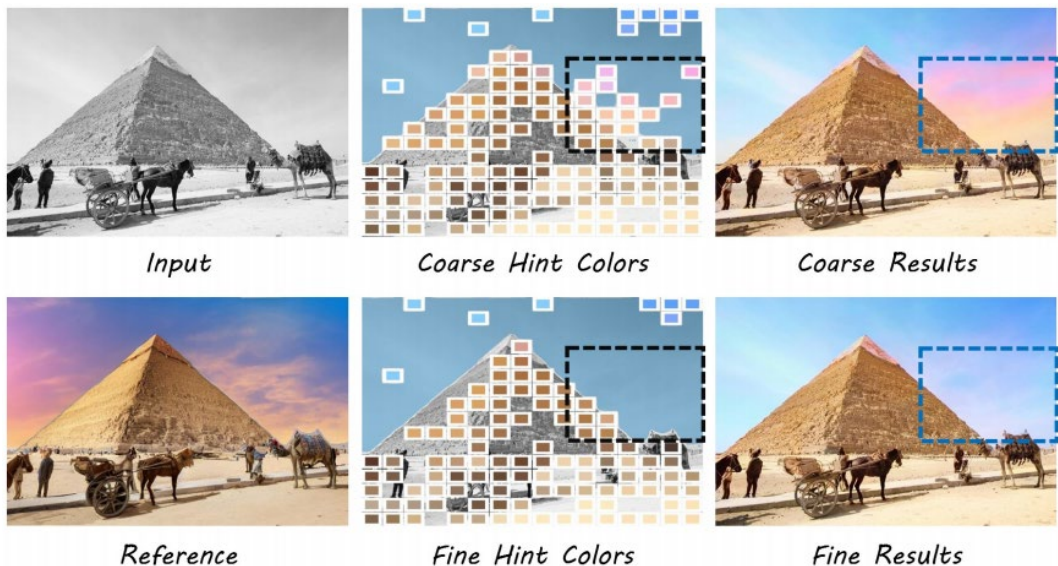


图7: 粗到细的提示颜色优化

- 通过语义对应关系生成初始的粗提示颜色集，但可能存在语义颜色模糊问题（如框选区所示）。
- 通过优化得到细提示颜色集，减少颜色溢出并确保上色结果更自然。

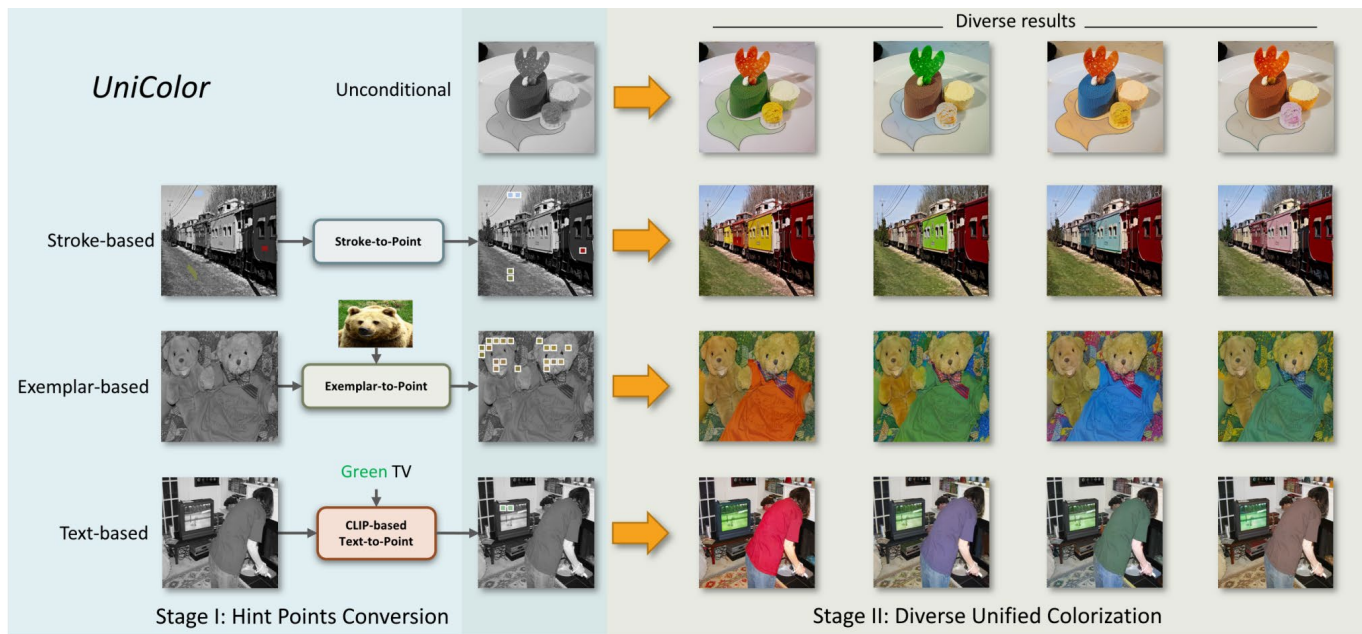


Fig. 2. Our unified colorization pipeline. The pipeline consists of two stages. In the first stage, all different conditions are unified as hint points. In the second stage, diverse results are generated automatically either from scratch or based on the condition of hint points. Input images: the 1st row is from ImageNet and all others are from MSCOCO.

上色模块(Colorization) 受到 统一上色框架 UniColor 的启发。UniColor是一个基于多模态条件的统一上色框架。

- 多模态条件，如笔画、参考图像和文本，被转换为提示颜色，这些颜色作为后续上色的指导。
- 上色过程分为两个步骤：从粗到细的提示颜色生成和提示颜色传播。

多模态的自动图像上色

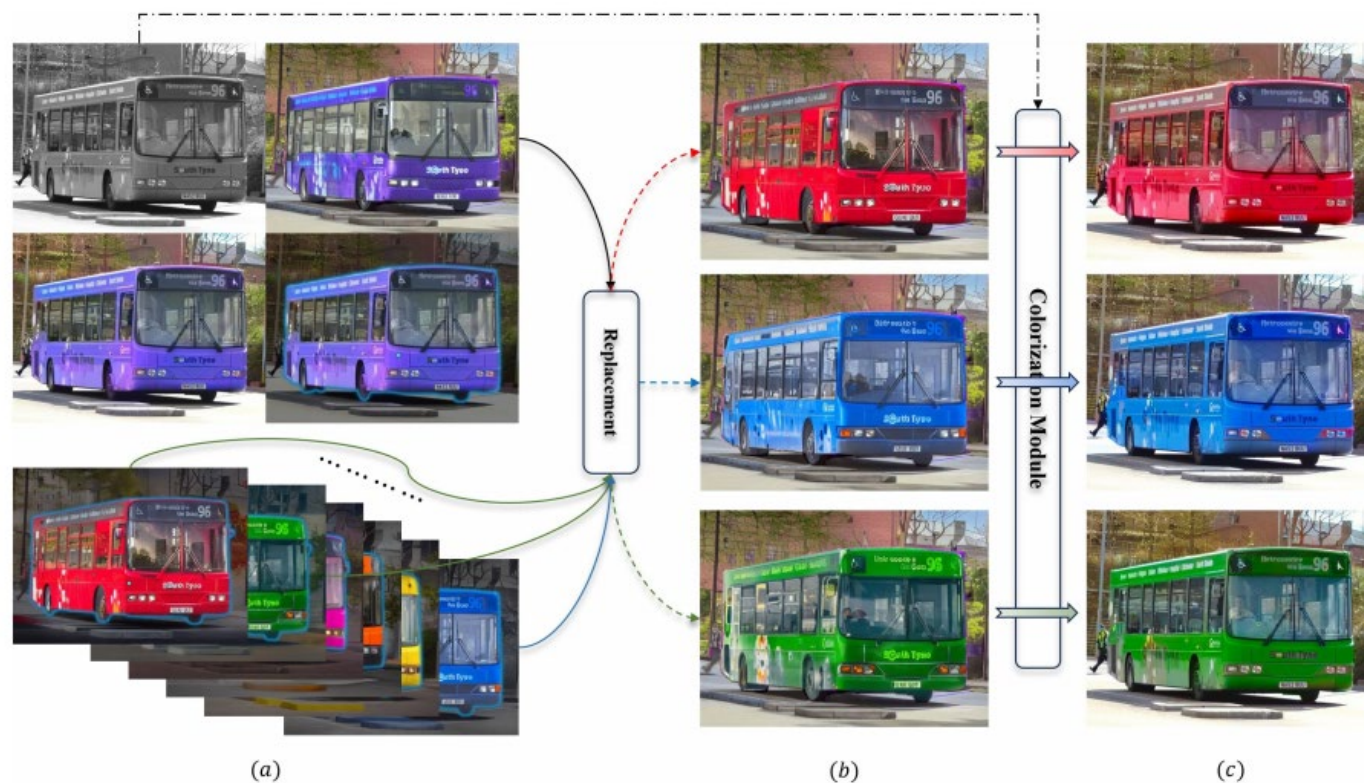


Figure 6. **User Interactive.** The top panel of (a) shows the black-

图6:

- (a)上方为黑白输入图像、参考图像、上色结果以及**用户标记不满意区域**；下方为**想象模块**生成的多个参考候选图像。
(b)用户选择了一个更满意的参考候选图像段来**替换**原有参考图像中的不满意部分，形成**新的参考图像**。
(c)展示了基于新参考图像调整后的上色结果，体现了框架的可编辑性和用户交互性。

多模态的自动图像上色

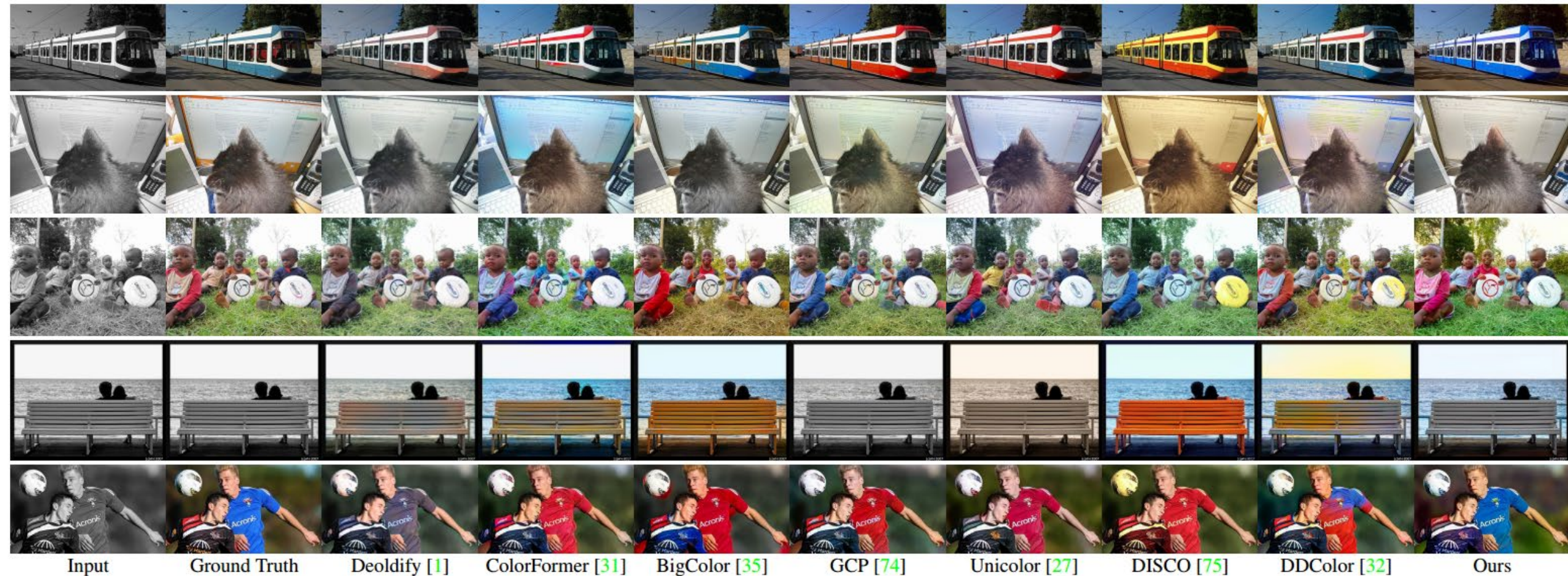


Figure 8. **Qualitative comparisons on the COCO validation set.** Our method can generate more natural and photo-realistic colors than state-of-the-art approaches. Zoom in for the best view.

多模态的自动图像上色

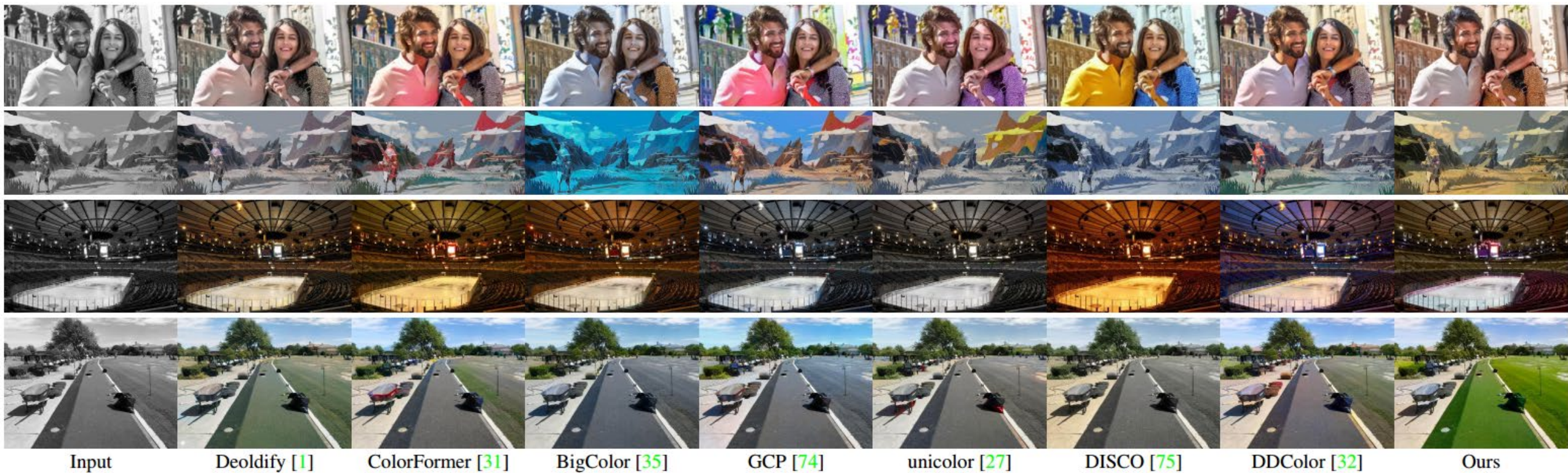


Figure 9. **Qualitative comparisons on in the wild validation set.** Our method can generate significantly more natural and photo-realistic colors than state-of-the-art approaches. Zoom in for the best view.

作业13

- 1、阅读1篇相关文献，撰写阅读报告
- 2、下载彩色化处理的算法代码，尝试阅读代码，并运行。(选做题)
 - ① 解读关键代码
 - ② 查看实验结果



THANKS
